



Good

Evening

Everyone ☺

- ✓ String Basics
- ✓ Change case of every character
- ✓ Check if substring $[i, j]$ is palindrome
- ✓ Length of longest palindromic substring
- ✓ Immutability - Memory View



STRING

A string can be defined as sequence of characters or in other words array of characters

"Welcome to Scala"

"Hello World"

Character

Single symbol in a single quote

'A' → 65

'B' → 66

'a' → 97

'A' to 'Z' → 65 to 90

'a' to 'z' → 97 to 122

'0' to '9' → 48 to 57

✓ `char ch = (char) 65;`

`print(ch)` $\Rightarrow$ 'A'

✓ `char ch = (char) ('a' + 1)`

`print(ch)` $\rightarrow$ 'b'

✓ `int x = 'a';`

`print(x)` $\rightarrow$ 97

Question 1. Togglecase

Given a string consisting of only alphabets (either lowercase or uppercase) print all characters of string in such way that for all lowercase character, print its uppercase and for all uppercase character, print its lowercase character

Input : "Hello"

hELLO

Quiz.

String = "aDgBHJe"?
AdGBhjE

Code:

```
function solve (String A) {
```

```
    N = A.length();
```

```
    for (i = 0 to N-1) {
```

```
        if (A[i] >= 'a' && A[i] <= 'z') {
```

```
            A[i] = (char) (A[i] - 32)
```

```
        }
```

```
        else if (A[i] >= 'A' && A[i] <= 'Z') {
```

```
            A[i] = (char) (A[i] + 32)
```

```
        }
```

```
    }
```

```
    print (A);
```

```
}
```

X

Strings are immutable

so you cant modify at a particular
index.

CHANGING SAME STRING IS NOT POSSIBLE IN JAVA & PYTHON

function solve (String A) {

String res = ""

for (i → 0 to A.length() - 1) {

char ch = A.charAt(i)

if (ch >= 'a' && ch <= 'z') {

res = res + (char) (ch - 32)

}

elseif (ch >= 'A' && ch <= 'Z') {

res = res + (char) (ch + 32)

}

}

print (res)

}

abc

abcd

abcdef

Using Character array instead

String solve (String A) {

char [] arr = A.toCharArray();

for (int i=0; i < arr.length; ++i) {

if (arr[i] >='A' && arr[i] <='Z') {
arr[i] = (char) (arr[i] + 32);

}

else if (arr[i] >='a' && arr[i] <='z') {

{

arr[i] = (char) (arr[i] - 32);

}

}

return new String (arr);

}

PYTHON :

class Solution :

def solve (self , A) :

char-list = []

for c in A :

if 'A' <= c <= 'Z' :

char-list.append (chr (ord (c) + 32))

else

char-list.append (chr (ord (c) - 32))

return ' '.join (char-list)

Quiz 2.

Considering the immutability of strings in most prog. languages, if you start with an empty string and keep adding one character at a time, how many string objects will be created when forming the string "abcdef"?

It

$$\frac{6}{7}$$

Substring

A substring is a contiguous sequence of characters within a string
A substring concept in string is similar to subarray concept in array

- ✓ Continuous part of string
- ✓ Full string can be substring
- ✓ A single char can be a substring

① "abc"

a	b
ab	bc
abc	c

Quiz 3.

String = "bxcd"

↓
b x c d
bx xc cd
bxc xcd
bxcd

$$4 + 3 + 2 + 1 = 10$$

$$1 + 2 + 3 + 4 \dots n$$

$$\frac{n(n+1)}{2} \cdot \frac{4(5)}{2} = \frac{2010}{2}$$

Question.

CHECK PALIDROME

Check whether the given substring of string s is palindrome or not.

A palindrome is sequence of characters that reads the same forward & backward. Example $\rightarrow$

"nayan"

"madam"

Input:

$s = \text{"ana**madam**spe"}$

start = 3

end = 7

True.

code

function ispalindrome (character array s[], start, end) {

while (start < end) {

if (s[start] != s[end])

return false;

else {

start++

end--

return true

TC :- $O(n)$
SC :- $O(1)$

Question 3

Longest Palindromic Substring

Given a string s , calculate the length of longest palindromic substring in s .

"ana **madam** m"

↓
5

Quiz

"feacabacabgf"

⇒ 7.

longest = 1 2 3 4 5 6 7 8 9

Quiz.

"a d a e b c d f d c b e t g g t e"

e b c d f d c b e

→ odd

e t g g t e

→ even.

BRUTE FORCE

function longestPalindrome (character array S[]) {

N = S.length
ans = 0

TC: $O(N^3)$
SC: $O(1)$

for (i = 0 to N-1) {

for (j = i to N-1) {

start = i

end = j

if (isPalindrome(S, start, end))

int length = end - start + 1

if (ans < length)

ans = length

}

print (ans)

OPTIMISED APPROACH

$\rightarrow 1 \times 2 \times 5$

"abc~~d~~edc"

$\Rightarrow 5$

$l = 1 \times 2 \times 4$

"x~~b~~aaby"

$\Rightarrow \underline{4}$

length = ~~1~~ ~~2~~ ~~4~~
5

"abb~~a~~bb"

code:

function longestPalindrome (character array s[])

maxlength = 0

N = s.length

TC : $O(n^2)$

SC : $O(1)$

for (c → 0 to N-1) {

// odd length center

left = c

right = c

while (left ≥ 0 && right < N) {

if (s[left] != s[right]) {
break;

}

left--;

right++

}

int length = right - left + 1

if (length > maxlength)

maxlength = length

}

// even length

left = c

right = c + 1

while (left >= 0 and right < N) {

if (s[left] != s[right])

break;

}

left--
right

}

int length = right - left + 1

if (length > maxlength)
maxlength = length

}

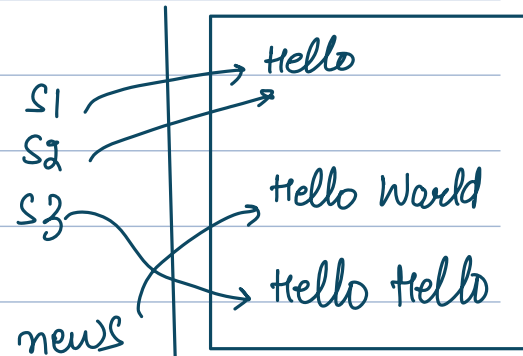
print (maxlength)

}

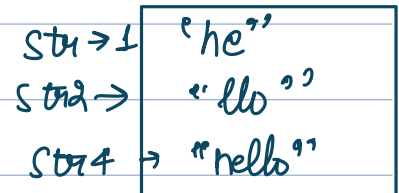
STRING IMMUTABILITY

```
String s1 = "Hello";  
String s2 = "Hello";  
String s3 = "Hello";  
s3 = s1 + s2;    => HelloHello
```

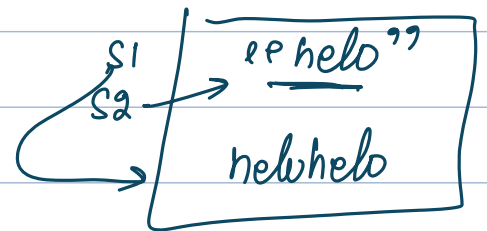
```
String newS = s3.concat("World");
```



```
str1 = "he"  
str2 = "llo"  
str4 = str1 + str2
```



```
s1 = "helo"  
s2 = s1
```



```
s1 = s1.concat(s2)
```

"helo helo"

str 1 = "he"

str 2 = "llo"

str 3 = "hello"

str 4 = str + str 2

hello

